SHAGGAR INSTITUTE OF TECHNOLOGY
Excellence in Science and Technology

Advanced Physics (Calculus-Based) - Course Syllabus

1. Basic Course Information

Course Code and Title:

Phys 1410- Introductory Physics II

Credit Hours:

Instructor(s):

Lectures by Solomon Bililign, Discussion and recitation, and breakout sessions

Dr. Nebiyu Gemechu

Ato Tesfaye Mamo

Semester/Quarter:

2nd Quarter

Prerequisites:

Phys 1310

Co-requisites:

Class Schedule:

Tuesday and Thursday, 2 PM to 3:15 PM Addis Ababa Time

Virtual Office Hours:

An office hour with Bililign: Mondays 6-8 PM Addis Ababa Time on Zoom for any student who needs discussion on topics related to the course

https://ncat.zoom.us/j/8468766150

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Contact Information:

Bililignsol@gmail.com

2. Course Description

Provide a brief description of the course, including scope, purpose, and relevance to the program.

Calculus-based introduction to classical electricity and magnetism, including such topics as, electric charge and electric fields, Gauss's law, electric potential, capacitance, current, resistance, and circuits, magnetic fields, and fields due to currents, induction and inductance, magnetism of matter, Maxwell's equations, and electromagnetic oscillations the basic concepts of wave motion, interference phenomena, electromagnetic waves: Poynting Vector, polarization and reflection, geometrical optics: mirrors, refraction, lenses, optical instruments, interference and diffraction.

3. Course Objectives- Learning objectives

List the broad goals of the course and how they align with the program educational objectives.

- Apply Coulomb's Law and superposition principle to calculate electric field due to a continuous charge distribution (uniformly charged line segment, circular or square loop, sphere, etc.)
- Apply Gauss' Law to compute electric field due to symmetric charge distribution
- Calculate electric field from electric potential and vice versa
- Compute the potential of a localized charge distribution
- Determine the surface charge distribution on a conductor in equilibrium
- Find the location and amount of all bound charges in a dielectric material
- Understand electric circuits and use Kirchhoff's laws in circuit analysis.
- Apply Biot-Savart Law and Ampere's Law to compute magnetic field due to a current distribution
- Explain dipoles – as point charges, and as generalized dipole vectors – for simple charge configurations.

- Explain similarities and differences between a conductor and a dielectric (both shield E, conductor shields E completely, dielectric shields via fixed dipoles, conductor shields via mobile electrons).
- Evaluate the torque or the force on a magnetic dipole in a magnetic field
- Explain the difference between para, dia, and ferromagnets, and predict how they will behave in a magnetic field.
- Use Faradays law to calculate electromotive force
- Calculate inductances of different systems
- Calculate energy in magnetic fields
- Calculate reflection and transmission of electromagnetic waves in linear media and relate the ideas to optics

4. Student Learning Outcomes (ABET Outcomes 1–7)

Upon successful completion of the course, students will be able to:

1. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. Apply engineering design to produce solutions that meet specified needs with consideration for public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. Communicate effectively with a range of audiences.
4. Recognize ethical and professional responsibilities in engineering situations and make informed judgments.
5. Function effectively on a team to provide leadership, create an inclusive environment, establish goals, plan tasks, and meet objectives.
6. Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. Acquire and apply new knowledge as needed, using appropriate learning strategies.

5. Mapping to Program Outcomes

Insert a table mapping each course learning outcome to the corresponding ABET student outcomes (1–7).

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6. Required Textbooks and References

List the required and recommended textbooks, manuals, and reference materials.

Physics for Scientists and Engineers with Modern Physics, 10th Edition

Serway and Jewett,

7. Course Topics and Schedule

Provide a week-by-week or session-by-session outline of topics, labs, assignments, and assessments.

THE WEEK OF MM/DD/YY	SUBJECT	UNIT LEARNING OUTCOMES (ULO)	HOMEWORK, EXAM Textbook chapters
Week 1 Feb 16	Electric Fields	• Electric charge and forces between point charges-Coulombs law • Superposition principle and calculating force due to discrete charge distribution • Define electric field • Use of a positive test charge to trace electric lines of force • Relationship between lines of force and electric field strength • Calculation of electric field of point charges • Calculation of electric field due to a continuous charge distribution	Physics for Scientists and Engineers with Modern Physics, 10th Edition Chapter 21 plus notes **Assignment 1-**

Week 2	Continuous Charge Distributions and Gauss's Law and electric potential	• Calculation of electric field due to a continuous charge distribution • Define electric flux and Gauss's Law • Express Gauss's law in differential form • Use gauss's law to calculate electric field of charge distributions • Use of symmetry • Express Gauss's law in differential form • Explain electric potentials and calculate electric potentials for discrete and continuous charge distributions • Explain electric potentials from electric fields and vice versa •	Physics for Scientists and Engineers with Modern Physics, 10th Edition Chapter 22 ,24 **Assignment 2-**
Week 3	Electric Potential Dipole potentials Advanced methods, dielectric materials	• Evaluate work and energy in electric fields	Physics for Scientists and Engineers with Modern Physics, 10th Edition Chapter 24, notes **Assignment 3-**

Week 4	More Capacitance and Dielectrics	• Definition of Capacitance • Calculating Capacitance • Combinations of Capacitors • Energy Stored in a Charged Capacitor • Capacitors with Dielectrics • Electric Dipole in an Electric Field • An Atomic Description of Dielectrics	Physics for Scientists and Engineers with Modern Physics, 10th Edition Chapter 25 **Assignment 4-**
Week 5-	Current and Resistance Direct-Current Circuits	• Electric Current • Resistance • A Model for Electrical Conduction • Resistance and Temperature • Superconductors • Electrical Power • Electromotive Force • Resistors in Series and Parallel • Kirchhoff's Rules • RC Circuits • Household Wiring and Electrical Safety	Physics for Scientists and Engineers with Modern Physics, 10th Edition Chapter 26 and 27 **Assignment 5-**

Week 6	Magnetic Fields Sources of the Magnetic Field	• Describe magnetic fields and Lorentz forces • Explain vector products to calculate magnetic forces • Evaluate forces due to current carrying wires • Explain line surface and volume currents • Discuss the sources of magnetic fields • Force of a magnetic field on a moving charge • Forces on current carrying wires and forces between current carrying wired • Introduction of Biot-Savart Law • Application to a line charge • Applications to current configurations and geometries • Applications to current loops, solenoids and Toroid's • Evaluate the divergence and curl of magnetic fields • Describe Ampere's Law • Explain Ampere's Law • Express Ampere's law in integral and differential form	Physics for Scientists and Engineers with Modern Physics, 10th Edition Chapter 28, 29 **Assignment 6-** **Complete:** Exam 1 Chapters 21-27

Week 7	Faraday's Law Inductance.	• Faraday's Law of Induction • Motional emf • Lenz's Law • The General Form of Faraday's Law • Generators and Motors • Eddy Currents Self-Induction and • Inductance • RL Circuits • Energy in a Magnetic Field • Mutual Inductance • Oscillations in an LC Circuit • The RLC Circuit	Physics for Scientists and Engineers with Modern Physics, 10th Edition Chapter 30 and 31 **Assignment 7-**
Week 8	Electromagnetic Waves	• Displacement Current and the General Form of Ampère's Law • Maxwell's Equations and Hertz's Discoveries • Plane Electromagnetic Waves • Energy Carried by Electromagnetic Waves • Momentum and Radiation Pressure • Production of Electromagnetic • Waves by an Antenna • The Spectrum of Electromagnetic Waves	Physics for Scientists and Engineers with Modern Physics, 9th Edition Chapter 33 **Assignment 8- Exam 2 (Chapters 28-33)**

Week 9	The Nature of Light and the Principles of Ray Optics Image Formation Wave Optics Diffraction Patterns and Polarization	• The Nature of Light • The Ray Approximation in Ray Optics • Huygens's Principal Dispersion • Total Internal Reflection • Images Formed by Spherical Mirrors • Images Formed by Thin Lenses • Optical Instruments • Young's Double-Slit Experiment • Analysis Model: Waves in Interference • Intensity Distribution of the Double-Slit Interference Pattern • Change of Phase Due to Reflection • Interference in Thin Films • The Michelson Interferometer	Physics for Scientists and Engineers with Modern Physics, 9th Edition Chapter 34,35 **Assignment-9-**
Week 10- April 26	Diffraction polarization and wave Optics	• Introduction to Diffraction Patterns • Diffraction Patterns from Narrow Slits • Resolution of Single-Slit and Circular Apertures • The Diffraction Grating • Diffraction of X-Rays by Crystals • Polarization of Light Waves • Wave optics	Physics for Scientists and Engineers with Modern Physics, 10th Edition **Chapter 36, 37** **Assignment 10** **Exam 3 (Chapter 34-37)**

			Final exam- Comprehensive Final project presentation

8. Laboratory Component

Describe lab activities, equipment used, safety requirements, and expected outcomes.

9. Grading Policy

Breakdown of grades: assignments, quizzes, lab work, midterms, final exams, participation, etc.

Category	**# of Activities**	**Percentage Grade Weight**
Assignment	10	20%
Exams	4	60%
Project	1	20%
Total		**100%**

Grading Scale

94% and above	A	83% - 80%	B	69% - 67%	C-
93% - 90%	A	79% - 77%	B-	66% - 60%	D
89% - 87%	A-	76% - 74%	C+	59% - 0%	F
86% - 84%	B+	73% - 70%	C		

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10. Academic Integrity

State the academic honesty policy and consequences in accordance with institutional regulations.

For SIT's policies about Academic Integrity, Class Attendance, Classroom Behavior, Accessibility and Student Support, and Safety Considerations, please read the SIT Student Manual.

In particular the following three policies are essential for this course.

Academic Honesty: Cheating on a test or assignment seriously undermines the integrity of the academic system and will not be tolerated. Although a student is not cheating, he or she is expected to refrain from actions that could be suspicious. Using common sense on your part should avoid unnecessary embarrassment.

Attendance: Attendance will be taken at every meeting. Students are expected to attend every online class and to be on time. If you are absent from any online and/or in person class for any reason, it is your responsibility to find out what you missed, including any announcements.

Classroom Behavior: You are expected to be in the classroom before the start of class time and leave at the end of the class time. Excessive talks without the permission of the instructors is prohibited. The use of any electronic device is allowed for the lecture and class activities only.

11. Accessibility and Student Support

Information on disability services, tutoring, counseling, and academic support resources.

12. Safety Considerations

Outline safety guidelines for labs, fieldwork, or other activities.